Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

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Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of four post-denoising corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming. Generation is driven by a reference-parity multi-modal guider (split per-modality classifier-free guidance, modality-isolation guidance, and spatiotemporal guidance via a value-passthrough self-attention skip) that correctly handles ComfyUI's packed audio-video latent representation, and all streaming noise is seeded for bit-reproducible runs. A two-stage pipeline generates a draft audio-video via CLSS streaming and then refines it using a 3-step distilled-LoRA schedule at $2\times$ spatial resolution.

We report matched text-to-video (T2V) and image-to-video (I2V) experiments: single-scene, ≈ 53-second clips generated as ten chunks (new_lf $= 16$, overlap $= 8$, per-chunk phase-jittered) under an identical seed and guidance stack (split per-modality CFG 4.0 video / 7.0 audio, STG 1.0 on both modalities at block 28, rescale 0.70; three transformer passes per step), differing only in the presence of a guide image. Stage 1 holds video boundary cosine similarity on a rising trajectory in I2V ($0.79 \to 0.98$) and a flat first-to-last trajectory with a mid-run floor in the dynamic-content T2V run ($0.796 \to 0.790$, floor 0.469

at chunk 5); a two-band spatial detail anchor holds the high-frequency energy share flat across the full run in both modes ($0.509 \to 0.509$ I2V, $0.486 \to 0.486$ T2V), closing the progressive-softening channel, and a reference-exact RMS anchor holds audio loudness to its chunk-1 level throughout ($\pm 5\%$). The per-chunk loudness-envelope repetition (the "metronome" of earlier chunked-audio runs) is diagnosed as a *fixed point of the constant-shape chunk grid*: with a constant overlap, every continuation chunk denoises an identical window and keeps the same phase of the model's window arc, so the same loudness gesture recurs each chunk. It is eliminated chunk-natively by a per-chunk overlap phase-jitter — cycling the overlap through [full, half, $\frac{3}{4}$] of its configured value so no two consecutive chunks share a window shape — which preserves arbitrary-length streaming and introduces no audio/video offset; with the phase lock broken structurally, the previously harmful audio SLB is restored as the content-continuity carrier and the RMS anchor keeps energy stability. Against earlier matched runs, the loudness-envelope correlation drops from a sustained 0.76–0.96 to ≤ 0.19 at every hand-off (-0.51–0.19 across both runs), audio boundary similarity rises from 0.07–0.53 to a 0.85–0.92 typical band ($0.619 \to 0.918$ first-to-last in I2V, $0.849 \to 0.916$ in T2V), and both phase-lock detectors read fully scattered (peak mode covering 1/10 chunks in each run). Audio high-frequency energy growth, previously unbounded, is closed in T2V ($0.885 \to 0.772$) and reduced to a mild residual in I2V ($0.881 \to 1.047$). Stage 2 ($2\times$ resolution, 3-step distilled refinement in four windows) refines video against frozen Stage-1 audio, with its own per-window detail anchor referenced to the upscaled Stage-1 slice; end-of-window fidelity to Stage-1 structure stays at 0.91–0.97 (I2V). Total generation time is $\approx$ 96–97 minutes per clip on a single 16 GB GPU.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip (81,840 tokens at 22×40 spatial) into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying five lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Five closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor that closes the

progressive-softening channel the one-sided shrinkage leaves open (Sections 3.1–3.5).
2. A per-chunk *overlap phase-jitter* mechanism that breaks the fixed-point repetition of chunked autoregressive audio at the chunk-grid level — cycling the per-chunk overlap through [full, half, $\frac{3}{4}$] of its configured value so no two consecutive chunks share a window shape — together with the audio SLB it makes safe to restore as the content-continuity carrier, all while preserving arbitrary-length streaming and exact audio/video proportionality (Section 5.4).
3. A linear stability analysis of the feedback loop in latent space, including an empirical per-chunk gain measurement protocol (Section 3.6).
4. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
5. A reference-parity multi-modal guider for ComfyUI (split per-modality CFG, modality-isolation guidance, and STG via value-passthrough self-attention skip) that correctly handles ComfyUI's packed audio-video latent representation (Section 5.6).
6. ComfyUI custom nodes implementing the full pipeline with seeded, bit-reproducible streaming noise and per-chunk reproducibility telemetry: unconditional settings dumps, per-frame origin/layout drift event tables, and a phase-lock repetition detector (Sections 5.5, 7.6).
7. Matched ≈ 53-second T2V and I2V generations, identical seed and guidance stack, streamed as ten chunks each, isolating the effect of visual conditioning on continuity and drift, and resolving the loudness-envelope repetition earlier runs exhibited: diagnosed as a fixed point of the constant-shape chunk grid — after being shown immune to every conditioning- and guidance-level intervention, which is exactly what motivated the grid-level diagnosis — and eliminated by the overlap phase-jitter plus the restored audio SLB (Sections 6, 5.4, 7.6).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an 8× temporal downscale (video VAE, causal) and a 160 × 4× audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160 \times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N, the last overlap_lf latent frames from chunk $N-1$ are prepended to the new chunk's latent, and the denoiser is conditioned on them at noise level τ_c:

$$\tilde{L}_{\text{overlap}} = (1 - \tau_c)\, \hat{L}_{\text{overlap}} + \tau_c\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{1}$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX's framework this is implemented as `VideoConditionByLatentIndex` with strength $= 1 - \tau_c$.

Implementation note: masked timestep, not a one-shot draw. Eq. (1) describes the target distribution of the overlap context, but the sampler does not literally draw ε once and hand the model a truncated schedule from τ_c. Tracing the actual mechanism (`ComfyBaseModel.process_timestep` for the LTX model, which the `noise_mask/strength` produced by `VideoConditionByLatentIndex` feeds into) shows that the per-token timestep supplied to the transformer at every sampling step t is $\tau_{\text{eff}} = (1 - \text{strength}) \cdot \sigma_t = \tau_c\, \sigma_t$ for overlap tokens, versus σ_t for the rest of the sequence — a *persistent* per-step rescaling of the global noise schedule, applied at every step, not a single noise-injection event applied once before an otherwise-standard remaining schedule. For a *linear* (rectified-flow) interpolation path, $x_t = (1-t)x_0 + t\varepsilon$, these two descriptions coincide exactly: scaling every step's effective time by a constant factor τ_c is algebraically identical to drawing x_{τ_c} once via Eq. (1) and continuing the *same* linear ODE from τ_c down to 0. LTX-2.3 is trained with a rectified-flow objective (Section 2), so the equivalence holds for this model; it would not hold in general for a non-linear (e.g. cosine or EDM) noise schedule, where per-step time-rescaling and one-shot truncated-schedule sampling are not interchangeable.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \tag{2}$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the "on-manifold" latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^*, then $\|\delta_{N+1}\| \approx g\|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \to \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.6 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds five closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1 - \beta) \cdot (1 - \tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1-\lambda)\,\mu_c^{(N-1)} + \lambda\,\hat{\mu}_c^{(N)} \tag{3}$$

$$\sigma_c^{(N)} = (1-\lambda)\,\sigma_c^{(N-1)} + \lambda\,\hat{\sigma}_c^{(N)} \tag{4}$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N's new frames. After generation, the overlap region is dropped and *only the new frames* L_N^{new} (Algorithm 1, line 5) are blended toward the EMA reference via AdaIN with blend factor β:

$$\tilde{L}_N^{\text{new}} = (1-\beta)\,L_N^{\text{new}} + \beta\ \text{AdaIN}(L_N^{\text{new}} \to \mu^{(N-1)}, \sigma^{(N-1)}) \tag{5}$$

The overlap region itself is never re-touched by AdaIN — it was already corrected as *previous* chunk's L_{N-1}^{new} before being pushed to the SLB. This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift**: the EMA σ_c is clamped to at most $(1+\Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification**: per-channel upward scaling is capped at $1.2\times$ the current chunk's std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min\left(1,\ \left(\frac{E_{\text{ref},b}^{(c)}}{E_b^{(c)}}\right)^{\gamma_b}\right) \in (0, 1] \tag{6}$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c, $E_{\text{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\boldsymbol{\gamma} = (0.0,\ 0.05,\ 0.2,\ 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the *last* frame of chunk 1 (not the first) and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). Additionally, a *forced insertion* every F_a chunks (chunk_index mod $F_a = 0$, chunk 1 always seeds regardless) prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered. The reference default is $F_a = 5$; the ComfyUI node auto-derives $F_a = \max(2, \min(5, \lceil \text{num\_chunks}/4 \rceil))$, giving $F_a = 3$ for the ten-chunk experiments in Section 6.

Anchors are retrieved via cosine similarity and injected as `VideoConditionByKeyframeIndex` at frame index 0 of the current chunk. Anchors too similar to the current overlap (sim > 0.85) are skipped to avoid duplicating the SLB conditioning. LRU eviction discards the least recently used anchor when the bank is full.

3.5 Two-Band Spatial Detail Anchoring

The shrinkage of Section 3.3 is deliberately one-sided (Eq. (6) only ever attenuates), and this asymmetry has a measured long-run consequence: low-band energy inflates while high bands are only ever shrunk, so the *high-frequency share* of latent energy falls monotonically — +19.6% band-0 inflation over seven chunks in one measured run, with the uncapped band EMA itself drifting upward and legitimising the excess. Nothing in the shrinkage loop can *restore* band energy, and the perceptual result is progressive detail loss.

The detail anchor closes exactly this channel. Each frame is split into a spatial low band (a 3×3 average pool over the latent grid) and its high-band residual, and the mean squared energy of each band is equalised toward a *fixed* scene-first reference (captured at chunk 1, re-baselined on scene change) with symmetric per-chunk gains:

$$g_{\text{lo}} = \left(\frac{E_{\text{lo}}^{\text{ref}}}{E_{\text{lo}}}\right)^{1/2} \text{ clamped to } [0.90, 1.10], \qquad g_{\text{hi}} = \left(\frac{E_{\text{hi}}^{\text{ref}}}{E_{\text{hi}}}\right)^{1/2} \text{ clamped to } [0.90, 1.12] \tag{7}$$

Unlike the shrinkage EMA, the reference here does not track the output — drift cannot legitimise itself — and unlike the shrinkage gains, these are symmetric: lost high-band energy is restored, not merely excess removed. The per-chunk clamp bounds the correction so a genuine scene change cannot be fought harder than ±10–12% per chunk, and a ±0.5% dead-band makes the module a strict no-op when energies already match. The same mechanism runs per-window in Stage 2, referenced there to the upscaled Stage-1 slice of the same window (Section 5.2). In the reported runs it holds the Stage-1 high-frequency share flat end-to-end ($0.509 \to 0.509$ I2V, $0.486 \to 0.486$ T2V; Section 6).

3.6 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1-\beta)(1-\tau_c)$ before it reaches the next chunk's denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1-\beta)\cdot(1-\tau_c) \tag{8}$$

The combined closed-loop gain, accounting for the model's open-loop amplification g, is:

$$\rho_{\text{closed}} = g\cdot\rho_{\text{loop}} \tag{9}$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g.

Why the two factors multiply. $(1-\tau_c)$ and $(1-\beta)$ act on disjoint tensor regions at different points in the cycle — $(1-\tau_c)$ on the *overlap* region when it is *read* as context (Section 3.1), $(1-\beta)$ on the *new-frame* region when it is *written* as output (Section 3.2, and see the corrected notation there: AdaIN never re-touches the overlap). This is not two attenuators applied to the same signal at the same place; it is two attenuators the same error signal passes through exactly once per loop iteration: a perturbation in chunk N's output is written with gain $(1-\beta)$ (since AdaIN is applied uniformly across L_N^{new}, including the tail frames that become the next SLB), then read back with gain $(1-\tau_c)$ once it becomes chunk $N+1$'s overlap context. For a linear, time-invariant loop this is exactly the standard construction of a loop gain from attenuators in series — the product is valid *because* each attenuator sits on the one path the error must traverse per cycle, not because they act on the same coefficients. What Eq. (8) does not capture is any cross-term where the correction strength itself depends on chunk content (e.g. the AdaIN amplification cap, Section 3.2, engages non-uniformly); that content-dependence is exactly why g cannot be folded into a single constant either (below).

Note that Eq. (8) is a first-order approximation that treats the two corrections as independent multiplicative attenuators. In practice g is not constant: it depends on the step count, guidance strength, noise schedule, and the spectral properties of the current latent. The analysis therefore provides a stability condition rather than a precise quantitative bound. The frequency-band shrinkage (Section 3.3) provides additional spectral attenuation beyond ρ_{loop}, especially in high-frequency bands, but is not included in the linear gain model because its contribution is content- and chunk-dependent.

Hyperparameter Choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$, which empirically provides adequate margin given the band-energy-growth proxy for g discussed in Section 7 (note: not the g_{SLB} perturbation measurement below, which was not exercised in the reported runs). $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\boldsymbol{\gamma} = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-Chunk Gain Measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F = \epsilon \, \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta^{\text{SLB}}_{\text{output}}\|_F}{\|\delta\|_F} \tag{10}$$

where $\Delta^{\text{SLB}}_{\text{output}}$ is restricted to the last overlap_lf new frames—the portion that becomes the next chunk's SLB. This measures sensitivity of the boundary output to boundary input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain.

The `measure_g` config field existed from the start, but — caught only when this was checked directly rather than assumed — no code actually implemented Eq. (10) behind it; the flag was a documented no-op, hardcoded off with no node-level toggle. This has since been implemented: a second denoising pass with a perturbed overlap latent, holding the noise sample and every other conditioning input fixed, comparing the raw (pre-AdaIN/shrink) output tail against the unperturbed pass. It is verified at the tensor level — the perturbation is confined exactly to the overlap region, the norm-ratio formula recovers a known injected gain in a synthetic test — but **was not enabled in either experiment reported in Section 6**, both of which predate the implementation and ran with `measure_g=False` regardless; consequently no g_{SLB} value appears in any results table here, and the measurement itself is untested against a live transformer forward pass. The $g \approx 1.03$–1.05 figures in Section 7 are a *different*, informal proxy —the ratio of measured band-3 latent energy growth to its EMA reference—not an instance of Eq. (10). The two should not be conflated: the band-energy proxy reflects unshrunk spectral drift under the full correction stack as actually run, while g_{SLB} would isolate the transformer's own sensitivity to a boundary perturbation, holding every correction fixed. Executing `measure_g=True` on a future run is the direct way to close this gap; it is listed as an open item in Section 8.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \boldsymbol{\gamma}$; state: SLB, EMA, anchor bank

$L_{\text{ov}} \leftarrow \text{SLB.read()}$ ▷ may be empty for chunk 0
$\tilde{L}_{\text{ov}} \leftarrow (1 - \tau_c)\, L_{\text{ov}} + \tau_c\, \varepsilon$ ▷ Eq. (1); Sect. 3.1
$A \leftarrow \text{anchor\_bank.retrieve(top-}m\text{ by cosine sim)}$ ▷ Sect. 3.4
$L_N \leftarrow \text{Denoise}(\tilde{L}_{\text{ov}},\, A,\, \text{prompt}_N)$
$L_N^{\text{new}} \leftarrow L_N[\text{overlap\_lf} :]$ ▷ drop overlap region
$L_N^{\text{new}} \leftarrow \text{AdaIN\_blend}(L_N^{\text{new}},\, \text{EMA},\, \beta)$ ▷ Sect. 3.2
$L_N^{\text{new}} \leftarrow \text{band\_soft\_shrink}(L_N^{\text{new}},\, \text{band\_EMA},\, \boldsymbol{\gamma})$ ▷ Sect. 3.3
$L_N^{\text{new}} \leftarrow \text{detail\_anchor}(L_N^{\text{new}},\, E^{\text{ref}}_{\text{lo/hi}})$ ▷ Eq. (7); Sect. 3.5
$\text{EMA.update}(L_N^{\text{new}},\, \lambda)$; $\text{band\_EMA.update}(L_N^{\text{new}},\, \lambda)$
$\text{anchor\_bank.update(last frame of } L_N^{\text{new}})$ ▷ Sect. 3.4
$\text{SLB.push}(L_N^{\text{new}}[-\text{overlap\_lf} :])$
return L_N^{new}

5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. Dequantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.
2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only 2 blocks are resident on GPU at a time (≈ 5 GB model weights on GPU), with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap\_lf})$ instead of $O(\text{total\_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at 512×768 resolution:

Component	Peak VRAM
Model blocks (2 × BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, typically 11×20 for 704×512 output). Each chunk runs the full 20–30-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled-LoRA refinement). Stage 1 operates at 704×512 output resolution and 20 denoising steps, which is sufficient for temporal structure and scene content but produces visible quantisation artefacts and reduced texture detail, particularly after GGUF 4-bit dequantisation at the low-frequency latent scale. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid corresponding to 1408×1024 pixel resolution. Stage 2 then refines the full video in temporal chunks using a 3-step distilled-LoRA schedule with initial noise level $\sigma_0 \approx 0.909$. Each chunk's new frames are seeded from the upscaled Stage 1 latent with σ_0-scaled noise:

$$x_{\text{start}} = \sigma_0\,\varepsilon + (1 - \sigma_0)\,L_N^{\text{S1}} \tag{11}$$

This formulation anchors Stage 2 to the coarse temporal structure produced by Stage 1—preserving scene layout, motion trajectories, and chunk boundaries established by CLSS—while allowing the 3-step refinement to add local high-frequency detail. Stage 1 alone could in principle be generated at $2\times$ resolution, but this would quadruple token count and exceed VRAM budget under CPU block-streaming.

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- The detail anchor (Section 3.5) *does* run, per window, with the reference recomputed from that window's upscaled Stage-1 slice rather than carried from a first-window EMA. The 3-step distilled refinement systematically adds spatial high-frequency energy relative to its Stage-1 seed, and without the anchor this excess compounds across the window chain as end-of-run softening of the refined output relative to Stage-1 structure. Per-window re-referencing means the correction cannot accumulate: each window is equalised to its own source.
- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) to prevent grain seams at chunk boundaries—a single coherent noise tensor ensures that the per-chunk noise samples tile seamlessly when concatenated.
- Audio is *frozen*: the Stage-1 audio latent is passed through unchanged (denoise mask 0) and only video tokens are refined, with the frozen audio serving as cross-modal context. Re-rendering audio in Stage-2 windows was measured to re-seam it at every window boundary; freezing keeps the audio timeline bit-identical to Stage 1 (`frozen_verify_sim`=1.0 per window).

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of new_lf latent frames to $(\text{new\_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new\_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new\_af}/\big((\text{new\_lf} - 1) \cdot 8 + 1\big)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap\_af} = \text{round}(\text{overlap\_lf} \cdot 8 \cdot r) \tag{12}$$

With $\text{overlap\_lf} = 8$ and the 13-lf/102-af template this gives $\text{overlap\_af} = 67$. (An earlier per-chunk accounting ignored the causal first-frame discount, undercounting every continuation chunk by ≈ 0.28 s and desynchronising audio from video by ≈ 2 s over seven chunks; the pixel-time ledger above eliminates the drift exactly.) The effective video overlap is additionally clamped to new_lf: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Continuity via the audio SLB, restored. Audio continuity is carried by two complementary conditioning paths. The first is an audio analogue of the video SLB: the previous chunk's audio tail is seeded into the overlap region at τ_c, so the model *hears* what it continues rather than inferring it from context. The buffer is stored at the configured maximum length and each chunk slices the tail matching its jittered overlap (Section 5.4); the freeze is verified numerically every chunk (`SLB honored`= 1.0000 in both reported runs). The second path is a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from the previous chunk's pre-overlap tail and passed *clean and at full length* (67 af ≈ 2 s, decoupled from the video overlap; an implicit coupling to overlap_af starved it at small overlaps and measurably collapsed within-chunk audio coherence from 0.91 to 0.27). The SLB path has a confounded history worth recording. Under a *constant* overlap it was measured harmful and disabled: the buffer froze each chunk's loudest tail into the next chunk's opening, compounding into high-frequency runaway ($1.02 \to 1.51$ over ten chunks) and raw-RMS runaway that the energy anchor then fought with growing corrections. That left `ref_audio` as the only carrier, which proved far too weak alone — boundary cosine 0.07–0.53 in I2V, with every chunk re-inventing its audio and speech rendered unintelligible. In retrospect the SLB was blamed for a loop that lived in the chunk grid, not in the buffer: its "harm" was feeding the phase-locked fixed point analysed in Section 5.4. With the phase lock broken structurally by the overlap jitter, the same SLB shows no runaway (T2V `aud_hf` $0.885 \to 0.772$ over the run) and lifts boundary similarity to a 0.85–0.92 typical band. The division of labour is: **jitter = anti-repetition, audio SLB = content continuity, RMS anchor = energy stability**. Perturbing the reference window itself — blending in noise, shrinking its length over chunks, flattening its loudness envelope — was each tested live against the repetition and none helped (the energy-injecting variants added audible hiss); those failures were what redirected the search from conditioning *values* to the window *shape* (Section 7.6). The clean full-length reference remains the measured optimum.

Reference-exact RMS anchoring. Un-anchored streaming audio is a divergent loop: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio's global RMS toward an EMA target whose drift from the chunk-1 reference (computed onset-excluded) is hard-capped at ±5% (the same Δ_σ used for the video EMA), before the chunk's tail is fed forward as the next reference window. Anchoring *before* feed-forward is what keeps the recursion safe at any sequence length. In the T2V run this anchor absorbs raw per-chunk loudness excursions of 0.734–0.842 against a chunk-1 onset-excluded reference of 0.8404, holding output RMS at 0.806–0.840 (largest correction gain 1.12, chunk 3); in I2V it absorbs raw excursions of −9%/+19% (0.511–0.663 against a reference of 0.5589), holding output at 0.554–0.587 with corrections of up to 14% downward (Sections 6.2.3, 6.3.3). On multi-scene runs the anchor is re-baselined at scene changes; this is measurably a ratchet (each re-baseline legitimises accumulated loudness creep: 0.599 → 0.699 → 0.823 over three scenes in one run) and remains a known limitation.

Boundary smoothing. At each audio chunk boundary, a linear blend over ±2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between consecutive chunks, which would otherwise be visible as discontinuities in the spectrogram.

5.4 Per-Chunk Overlap Phase-Jitter

The loudness-envelope repetition analysed in Section 7.6 turned out to live neither in the conditioning values nor in the guidance weights, but in the *geometry of the chunk grid*. With a constant overlap, every continuation chunk denoises a window of identical shape: same overlap length, same total window length, hence the same split between the regenerated-and-dropped lead-in and the kept new frames. The kept region is therefore always the same phase of the model's window arc — its canonical loudness gesture for the static prompt — and the chunk map acquires a fixed point: each chunk re-renders the arc and hands the next chunk a context that elicits the same arc again. The lock engages exactly when the reference-audio window first reaches full length (chunk 2–3), which is why listeners hear the audio "start repeating after chunk 1". This reading also explains the intervention record of Section 7.6: every failed intervention modulated conditioning *values* inside a window whose *shape* it left untouched, and a fixed point of the window shape is invisible to all of them.

The fix is to make a fixed point impossible. The per-chunk overlap cycles through [full, half, $\frac{3}{4}$] of the configured value — $[8, 4, 6]$ latent frames for overlap = 8; the realised sequence in the reported runs is $4, 6, 8, 4, 6, 8, \ldots$ — so no two consecutive chunks share a window shape and no single arc phase can be kept twice in a row. Three invariants keep the scheme transparent to the rest of the pipeline:

- **Timeline invariance.** Every chunk still keeps exactly new_lf video and new_af audio frames; only the regenerated-and-dropped lead-in varies in length. The assembled output timeline is unchanged relative to a constant-overlap run.
- **A/V proportionality.** The audio overlap derives from the same per-chunk video overlap (34/50/67 af for 4/6/8 lf in the reported runs), so video and audio overlaps scale together and no audio/video offset accumulates.
- **Buffer slicing.** The video and audio SLBs are stored at the configured maximum overlap; each chunk slices the last N frames matching its jittered value, so buffer management is unchanged.

The jitter shortens the effective context on half-overlap chunks (4 lf); no video-continuity loss is measured at the ten-chunk horizon, but longer horizons are untested (Section 8). A structurally simpler alternative — re-rendering the audio of the assembled Stage-1 video in a single non-chunked pass — was implemented and worked for short clips, but was rejected by design: CLSS exists for arbitrary-length streaming, and a whole-video audio window reintroduces exactly the $O(\text{total\_frames})$ context scaling the chunking removes (at this model's joint audio-video token count it is also VRAM-blocked outright, $\approx$ 140k tokens against a 42k budget). The phase-jitter achieves the anti-repetition effect chunk-natively, at any length.

5.5 ComfyUI Node Integration

Five custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig**: exposes the CLSS hyperparameters worth exposing (τ_c, β, overlap); validated internals are fixed.
2. **CLSSScenePrompts**: multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks. With one prompt per chunk this doubles as a per-chunk *object-identity* anchor (see Section 8).
3. **CLSSStreamingSampler**: Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Implements the per-chunk overlap phase-jitter (Section 5.4) and emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector.
4. **CLSSStage2**: Stage 2 refinement sampler with consistent per-chunk noise slicing, per-window detail anchoring, and frozen-audio passthrough (Section 5.2).
5. **CLSSAVGuiderV2**: reference-parity multi-modal guider (Section 5.6). Its predecessor (**CLSSAVGuider**, a `sampler_cfg_function` patch) is retained but inert on current ComfyUI and warns accordingly.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.6 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently (`MultiModalGuider`):

$$\mathrm{pred} = c + (s_{\mathrm{cfg}} - 1)(c - u) + s_{\mathrm{stg}}\,(c - p) + (s_{\mathrm{mod}} - 1)(c - m) \tag{13}$$

per modality, followed by a rescale pred $*= r\,\sigma_c/\sigma_{\mathrm{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio↔video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection (out $= W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. The reference defaults are video cfg 4–4.5, audio cfg 7, modality 3, STG 1 for both modalities, rescale 0.7 (four transformer passes per step). The working configuration runs $s_{\mathrm{mod}} = 1$: the modality term of Eq. (13) vanishes and its pass is skipped (three passes per step). This is a measured choice, not an economy: raising modality guidance to the reference value 3 was tested live and made the audio audibly *worse* (hiss/drone; within-chunk audio coherence collapsed 0.68 → 0.61) while leaving the audio repetition it might have addressed unchanged — with Stage-2 audio frozen, the raw guided Stage-1 audio is the final audio, and additional guidance energy surfaces directly as audible high-frequency content.

The packed-latent representation. ComfyUI's `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v{+}N_a]$ tensor before sampling and unpacks only after. A guider that tries to detect the audio-video case with `isinstance(x, NestedTensor)` inside `predict_noise` therefore never sees a nested tensor and falls back to plain shared CFG — and it does so silently, producing no error and no log line. The guider must instead capture the per-modality shapes in `sample()` *before* packing, detect the packed $[B, 1, N]$ representation in `predict_noise`, unpack each pass output, apply Eq. (13) per modality, and repack. Two diagnostics guard against a silent regression: the guider announces the active branch and its pass count once per chunk (`passes/step=3` for the working configuration, `4` with modality guidance enabled), and it logs the STG difference norms so a zero-valued (inert) perturbation is visible immediately. A useful invariant follows from the pass count: because each guidance term costs a transformer pass, a guidance configuration that does not change per-step wall-clock time is not actually running — in an earlier 13-lf configuration the four-pass stack cost ≈ 17.3 s/it against ≈ 8.0 s/it for plain CFG. Two engineering rules generalise: guidance paths must announce which branch is active, loudly and unconditionally; and a guidance change that does not change step time did not happen.

6 Experiments

Two generation experiments share the same CLSS configuration, seed, and active guidance stack but differ in the first-frame conditioning: a text-to-video (T2V) run with no image prior (Section 6.2), and an image-to-video (I2V) run conditioned on a single guide frame ($1 \times 11 \times 20$ latent, Section 6.3). Both stream ten Stage-1 chunks (new_lf $= 16$, overlap $= 8$, jittered $[4, 6, 8]$ per chunk, Section 5.4) to 160 latent frames (≈ 53 s at 24 fps) and refine through the same four-window Stage 2 pass.

6.1 Shared Configuration

Table 1: Shared configuration. Guidance values are those actually active during sampling (verified by the guider's per-chunk `passes/step` and difference-norm telemetry, Section 5.6). The Stage-2 geometry is selected automatically from the VRAM token budget.

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF)
Resolution	704×512 (Stage 1), 1408×1024 (Stage 2)
τ_c	0.05
β	0.40
λ	0.10
Δ_σ	0.05
F_a	3 (auto, for 10 chunks)
$\boldsymbol{\gamma}$	(0.0, 0.05, 0.2, 0.3)
Detail anchor	on (Stage 1 scene-first ref; Stage 2 per-window ref)
Video guidance	CFG 4.0, STG 1.0
Audio guidance	CFG 7.0, STG 1.0
Modality guidance	1.0 (disabled; Section 5.6)
STG block / rescale	28 / 0.70 (three passes/step)
S1 noise temporal-corr mix	0.30 (video only)
Audio continuity	audio SLB at τ_c (auto→on) + `ref_audio` 67 af, clean
Overlap jitter	on; cycle $[8, 4, 6]$ lf (full/half/$\frac{3}{4}$ of 8)
Audio anchor	RMS-only, reference-exact $\pm 5\%$
Stage 1 steps	20
Stage 2 steps	3 (distilled), audio frozen (mask 0)
VRAM (GPU)	15.61 GB (RTX-class)
T2V first-frame	Text only
I2V first-frame	Guide image, latent $[1, 128, 1, 11, 20]$
Stage 1 (both)	10 chunks, new_lf $= 16$, overlap $= 8$, 1 scene
Stage 2 (both)	4 windows, fpc $= 40$, overlap $= 13$ (auto)

6.2 Experiment 1: Text-to-Video (T2V)

No image is provided; Stage 1 chunk 1 begins from pure noise conditioned on the text prompt alone. The clip is a single prompted scene streamed as ten chunks (new_lf $= 16$, overlap $= 8$, jittered $[4, 6, 8]$), assembling to 160 latent frames (≈ 53 s), under the identical seed and guidance stack as the I2V run (Table 1). The prompted content is highly dynamic: intra-chunk similarity (a chunk's first vs. last new frame) stays low-to-moderate (0.02–0.71) for the whole run (Table 2), so this run exercises the correction stack under continuous motion rather than a stable anchored scene.

6.2.1 Guidance Activity

Every chunk reports `[CLSS AVGuiderV2] active: video_cfg=4.0 audio_cfg=7.0 modality=1.0 stg=v1.0/a1.0@blk[28] rescale=0.70 passes/step=3`, so no step fell back to plain shared CFG. The guider's step-1 STG difference norms stay well away from zero on both modalities throughout — video 83.8–112.7, audio 20.3 at chunk 1 rising to 46–53 by the final chunks — confirming both STG perturbations are live on every chunk (a zero norm would mean an inert self-attention skip). The mirror-image chunk-1 outlier appears in I2V on the *video* norm instead (Section 6.3), where the guide image is the unmatched conditioning.

6.2.2 Stage 1 Results

Table 2: T2V Stage 1 per-chunk video metrics (10 chunks).

Chunk	Boundary sim↑	Intra sim	Identity sim	ρ_{loop}
1/10	N/A	0.0215	1.0000 (ref)	0.570
2/10	0.7957	0.5740	0.7957	0.570
3/10	0.9185	0.2679	0.6716	0.570
4/10	0.7828	0.3469	0.7828	0.570
5/10	0.4685	0.2626	0.5317	0.570
6/10	0.5408	0.3013	0.5852	0.570
7/10	0.8607	0.7100	0.8607	0.570
8/10	0.7298	0.5166	0.7298	0.570
9/10	0.7801	0.2692	0.5815	0.570
10/10	0.7900	0.5356	0.6698	0.570

Without a guide image, video boundary continuity is far more variable than I2V's: the seams range from 0.469 (chunk 5) to 0.919 (chunk 3), with the weakest hand-offs sitting *mid-run* (chunks 5–6, 0.469/0.541) and the final three seams holding 0.73–0.79. The first-to-last trend is flat (0.796 → 0.790), so this is boundary *variance*, not accumulated drift — the floor does not deepen with chunk index. Chunk 5's dip coincides with the first half-overlap window (4 lf context) following a full-overlap chunk: the jitter (Section 5.4) trades a shorter context on those chunks for its anti-repetition effect, and the cost shows up here as seam variance rather than as drift. Intra-chunk similarity stays low throughout (0.02–0.71), confirming sustained motion within every chunk. Origin similarity to the scene reference is contained over the run (0.161 → 0.158, range 0.107), and identity similarity to the anchor bank holds a

0.53–0.86 band with no monotonic decay. The largest origin-similarity drops still concentrate at the chunk 1→2 hand-off (deepest $\Delta = -0.406$ at $t = 5.3\,\mathrm{s}$, with the next three largest also in chunks 2–5), while the largest coarse-layout drops spread across chunks 1, 2, 3, 4, and 9 — consistent with a run whose content moves throughout rather than drifting from a fixed opening.

Table 3: T2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains ($g_{\mathrm{lo}}/g_{\mathrm{hi}}$, Eq. (7); 0.9000 is the per-chunk floor cap).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	−0.0197	−0.0213	0.9962	0.9950/0.9929	0.9716	1.0341
3	−0.0100	−0.0133	0.9985	0.9889/0.9865	1.0027	1.0167
4	−0.0040	−0.0379	0.9941	0.9638/0.9765	1.0404	0.9361
5	−0.0118	−0.0206	0.9942	0.9855/0.9965	1.0129	0.9845
6	+0.0025	−0.0128	0.9949	0.9784/0.9955	1.0483	0.9749
7	+0.0047	−0.0416	0.9930	0.9636/0.9929	1.0591	0.9176
8	+0.0002	−0.0337	0.9927	0.9761/0.9954	1.0525	0.9401
9	+0.0047	+0.0036	0.9962	0.9922/0.9966	1.0758	1.0025
10	+0.0033	−0.0485	0.9915	0.9641/0.9921	1.0137	0.9181

Every T2V continuation chunk except chunk 9 receives a *downward* std correction ($\Delta\sigma < 0$; chunk 9 is +0.0036), in contrast to I2V's near-neutral pattern (Table 7): without an image-conditioned first chunk to compress toward, each text-only chunk's raw std runs hot relative to the EMA reference and AdaIN consistently pulls it back down. Spectral shrinkage is correspondingly more active than in I2V (band-2 gains down to 0.9636 vs. I2V's ≥ 0.983). The detail anchor works visibly hard in this run: raw high-band energy overshoots its scene-first reference on most chunks from chunk 4 (the model keeps adding spatial detail energy under sustained motion), pulling g_{hi} down to 0.918–0.940 while g_{lo} boosts the low band by up to 7.6% — the net effect is a high-frequency *share* held exactly at its 0.4863 reference on every chunk, where the same one-sided shrinkage without the anchor produced a monotonic hf-share collapse (Section 3.5).

The anchor bank grows to six entries over the run (bank_frame_ids = [15, 47, 63, 95, 111, 159]). With new_lf = 16, chunk k's last frame lands at absolute index $16k-1$: $15, 31, 47, 63, \ldots$ for $k = 1, 2, 3, \ldots$ — so these six entries are chunks $\{1, 3, 4, 6, 7, 10\}$. Four of the six are the forced-insertion schedule exactly (chunk_index mod $F_a = 0$ with $F_a = 3$: chunk 1 always seeds, then chunks 4, 7, 10). Chunks 3 and 6 are *organic*, threshold-triggered commits: chunk 2's similarity to the seed anchor was already below threshold (last_frame_max_sim = 0.6679, streak= 1) and chunk 3 pushed the streak to $P = 2$; the same pattern repeats at chunks 5–6 (last_frame_max_sim = 0.6113 at chunk 5). Because the forced positions are fixed by chunk index alone, the two runs' forced entries coincide exactly despite entirely different video content; the organic commits are content-dependent, and here T2V commits twice organically while I2V commits once (chunk 3 in both; I2V's chunk-5 similarity, 0.8621, stays above threshold, so its bank stops at five entries, Section 6.3).

6.2.3 Stage 1 Audio

Table 4: T2V Stage 1 per-chunk audio metrics. RMS raw is the chunk's pre-anchor loudness; RMS out is after the reference-exact anchor (chunk-1 onset-excluded reference 0.8404). SLB hon. is the frozen audio-SLB verification (1.0 = the seeded tail survived denoising).

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim	SLB hon.
1	0.745	0.622	—	0.842	(ref)	N/A	—
2	0.867	0.770	−0.510	0.835	0.840	0.8488	1.000
3	0.695	0.826	−0.003	0.743	0.830	0.8693	1.000
4	0.417	0.381	−0.035	0.786	0.826	0.8993	1.000
5	0.653	0.757	−0.135	0.824	0.826	0.7569	1.000
6	0.637	0.715	−0.338	0.734	0.816	0.7243	1.000
7	0.780	0.765	−0.285	0.818	0.816	0.9000	1.000
8	0.632	0.559	−0.194	0.761	0.811	0.8514	1.000
9	0.648	0.705	−0.046	0.776	0.807	0.3915	1.000
10	0.730	0.741	−0.258	0.793	0.806	0.9162	1.000

The reference-exact RMS anchor holds output loudness at 0.806–0.840 against raw per-chunk excursions of 0.734–0.842 (largest correction gain 1.12, chunk 3) — the raw collapse-and-creep pattern of the pre-anchor runs is gone, and the anchor now trims single-digit percentages rather than rescuing a +72% drift. The headline change over every earlier configuration is the envelope row: env_corr never rises above −0.003 at any of the nine hand-offs (trend $-0.510 \rightarrow -0.258$, range 0.507), where the matched constant-overlap runs sustained 0.76–0.96 from chunk 4 onward. The energy-peak detector reads fully scattered (peaks at frames [38, 118, 127, 43, 101, 64, 26, 86, 49, 12], mode covering 1/10 within ± 1): no shared phase, no recurring gesture — the fixed-point mechanism of Section 5.4 is broken by construction, and the audible metronome with it. Boundary similarity with the restored audio SLB holds 0.72–0.92 on eight of nine seams, rising first-to-last ($0.849 \rightarrow 0.916$); the single weak seam (0.392, chunk 9) is flagged by the trend summary and is the honest residual of this run. The SLB freeze is verified numerically at every chunk (`SLB honored`= 1.0000 throughout).

6.2.4 Stage 2 Results

Table 5: T2V Stage 2 per-window metrics (4 windows). $g_{\mathrm{lo}}/g_{\mathrm{hi}}$ are the per-window detail-anchor gains (reference: the window's own upscaled Stage-1 slice); Aud_bnd is the frozen Stage-1 audio's own continuity at the window seam (Stage 2 never re-rolls audio).

Window	Bnd sim↑	Intra sim	Fid first	Fid last	Aud_bnd	$g_{\mathrm{lo}}/g_{\mathrm{hi}}$
1/4	N/A	−0.3224	0.9190	0.9669	N/A	1.036/0.900
2/4	0.9509	0.5568	0.9667	0.9277	0.7783	1.048/0.900
3/4	0.5629	0.2996	0.9280	0.9343	0.9001	1.036/0.900
4/4	0.7740	0.5562	0.8992	0.9230	0.8311	1.021/0.900

Window 1's intra similarity is *negative* (−0.322), unlike I2V's positive 0.728 for the same window: the dynamic T2V opening (Table 2, chunk-1 intra 0.022) carries into Stage 2 as a first window whose start and end frames barely correlate, whereas I2V's guide image keeps its first window internally coherent. The per-window detail anchor engages identically at every window: the 3-step distilled refinement adds spatial high-frequency energy over its Stage-1 seed at every window (pre-anchor hf share 0.46–0.50 against Stage-1 references of 0.32–0.35), and g_{hi} rides its 0.90 floor cap in all four windows, pulling the excess back toward the Stage-1 reference. Because the reference is recomputed per window, the correction cannot compound: fidelity to Stage-1 structure holds 0.90–0.97 throughout (fid_first 0.919 → 0.899, fid_last 0.967 → 0.923 first-to-last), with no end-of-run collapse. The audio row measures the *frozen* Stage-1 audio's own continuity at the three window seams (0.778/0.900/0.831, rising with the Stage-1 audio's own improved seams) — Stage 2 introduces no audio seams of its own (`frozen_verify_sim`= 1.0 at every window).

6.3 Experiment 2: Image-to-Video (I2V)

A single guide image is encoded to a latent of shape $[1, 128, 1, 11, 20]$ and provided as first-frame conditioning to Stage 1 chunk 1 (guide adherence 1.0000 measured at the placed frame). The clip is a single prompted scene streamed as ten chunks (new_lf = 16, overlap = 8, jittered $[4, 6, 8]$), assembling to 160 latent frames ($\approx 53\,\mathrm{s}$). All CLSS hyperparameters and the guidance stack are those of Table 1.

6.3.1 Guidance Activity

Every chunk reports `[CLSS AVGuiderV2] active: video_cfg=4.0 audio_cfg=7.0 modality=1.0 stg=v1.0/a1.0@blk[28] rescale=0.70 passes/step=3`, so no step fell back to plain shared CFG. The step-1 STG difference norms stay well away from zero on both modalities on every chunk — video 81.6–133.5, audio 4.9–25.0 — with chunk 1 the clear video-side outlier (203.7): the guide image is a hard constraint the self-attention-skipped prediction cannot honour, so the perturbation response is largest exactly where the conditioning is strongest.

6.3.2 Stage 1 Results

Table 6: I2V Stage 1 per-chunk video metrics (10 chunks). Boundary_sim compares the last overlap frame of the previous chunk against the first new frame; identity_sim compares the chunk against its nearest anchor-bank entry.

Chunk	Boundary sim↑	Intra sim↑	Identity sim	ρ_{loop}
1/10	N/A	0.4934	1.0000 (ref)	0.570
2/10	0.7900	0.6997	0.7900	0.570
3/10	0.8650	0.6752	0.7542	0.570
4/10	0.9586	0.8974	0.9586	0.570
5/10	0.9373	0.8498	0.9373	0.570
6/10	0.9404	0.8175	0.8776	0.570
7/10	0.9713	0.9053	0.9053	0.570
8/10	0.9505	0.8755	0.9505	0.570
9/10	0.9614	0.8622	0.8513	0.570
10/10	0.9814	0.7029	0.9204	0.570

Video boundary continuity holds between 0.790 and 0.981 across all nine chunk transitions and *rises* over the run — the strongest seam is the last one. The floor sits at chunk 2 (0.7900), the first SLB-conditioned hand-off, after which no seam falls below 0.86: the corrected loop does not compound error over this horizon, and the jittered context length (4–8 lf) does not measurably cost video continuity here. The largest structural events of the whole run still cluster at the chunk 1→2 seam ($t \approx 5.3$–6.3 s): the four largest per-frame origin-similarity drops land there (deepest $\Delta = -0.190$), the transition out of the guide-image-locked opening. Global origin similarity to the scene reference is contained over the full run ($0.327 \to 0.290$, range 0.084), in contrast to the monotonic walk-away drift seen at wider overlaps in earlier open-loop testing. Identity similarity to the anchor bank dips early (floor 0.754 at chunk 3) and recovers to 0.92, and intra-chunk similarity rises from 0.49 to a 0.82–0.91 plateau as the scene settles — a stable anchored scene, the complementary regime to T2V's sustained motion.

Table 7: I2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains (Band 0 is the DC band, $\gamma_0 = 0$, gain$\equiv 1$, omitted).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	−0.0128	−0.0166	0.9955	0.9828/0.9924	1.0013	0.9782
3	−0.0029	−0.0072	0.9969	0.9888/0.9926	1.0184	0.9815
4	−0.0052	−0.0121	0.9961	0.9874/0.9982	1.0120	0.9843
5	−0.0031	−0.0109	0.9949	0.9912/0.9978	1.0134	0.9894
6	−0.0019	−0.0086	0.9949	0.9907/0.9957	1.0152	0.9947
7	−0.0032	−0.0094	0.9944	0.9920/0.9950	1.0209	0.9875
8	−0.0025	−0.0211	0.9955	0.9860/0.9941	1.0027	0.9732
9	+0.0014	−0.0158	0.9964	0.9855/0.9940	1.0175	0.9686
10	+0.0002	+0.0054	0.9989	0.9938/0.9925	1.0413	0.9918

On this stable anchored scene the correction stack idles, as designed: AdaIN deltas stay within $|\Delta\mu| \leq 0.013$, $|\Delta\sigma| \leq 0.021$, every band gain is ≥ 0.983, and the detail-anchor gains never stray more than 4.1% (low band) / 3.1% (high band) from unity — an every-chunk trim, not a fight. The result is the flattest telemetry of either run: high-frequency share exactly flat at its 0.5090 reference on every chunk, std stable ($0.967 \to 0.969$). Contrast T2V (Table 3), where the same stack works measurably harder every chunk; the corrections scale with measured excess rather than applying a fixed schedule.

The anchor bank grows from 1 to 5 entries over the run at chunks $\{1, 3, 4, 7, 10\}$ (bank_frame_ids = [15, 47, 63, 111, 159]) — the forced-insertion schedule plus the single organic commit at chunk 3 derived in detail for T2V (Section 6.2). I2V's second organic opportunity does not trigger: chunk 5's similarity to the bank (0.8621) stays above the $\rho_a = 0.78$ threshold, so its bank ends one entry smaller than T2V's.

The ComfyUI sampler uses the bank for identity telemetry only: the retrieval-injection conditioning path of Section 3.4 is present in the reference library but not wired into this sampler, so the bank measures identity drift without correcting it. Two conditioning wirings have been tried and both failed (Section 8): the reference library's own `VideoConditionByKeyframeIndex` appends a separate token block and reproduces the audio corruption documented for the i2v guide path (Section 5.6); an in-place low-strength replacement at the first new frame instead produced visible content morphing, because anchor retrieval returns the *least*-similar non-redundant anchor and so drags each chunk back toward content the scene has already moved past. No safe anchor conditioning mechanism is currently known; the bank remains diagnostic-only.

6.3.3 Stage 1 Audio

Table 8: I2V Stage 1 per-chunk audio metrics. Within-chunk similarity is reported at the start and end of each chunk; env_corr is the loudness-envelope correlation with the previous chunk (> 0.7 flags a repeating gesture); RMS raw/out are pre-/post-anchor (chunk-1 onset-excluded reference 0.5589); SLB hon. is the frozen audio-SLB verification.

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim	SLB hon.
1	0.347	0.523	—	0.556	(ref)	N/A	—
2	0.482	0.842	0.060	0.511	0.554	0.6190	1.000
3	0.740	0.811	0.164	0.570	0.556	0.8879	1.000
4	0.919	0.886	0.190	0.581	0.558	0.9474	1.000
5	0.791	0.748	−0.081	0.663	0.569	0.8736	1.000
6	0.838	0.787	0.167	0.617	0.574	0.8882	1.000
7	0.828	0.720	−0.076	0.596	0.576	0.9554	1.000
8	0.790	0.875	−0.001	0.657	0.584	0.9130	1.000
9	0.917	0.920	−0.057	0.615	0.587	0.9104	1.000
10	0.732	0.710	−0.020	0.595	0.587	0.9183	1.000

Loudness is intrinsically stable in this run and the anchor enforces it: raw per-chunk RMS ranges 0.511–0.663 (−9%/+19% around the 0.5589 reference) and the anchored output holds 0.554–0.587, with the largest corrections *downward* (14% at chunk 5, where the raw chunk runs loud). The loudness-envelope correlation never crosses the 0.7 repetition flag — it never exceeds 0.190 at any hand-off (trend $0.060 \rightarrow -0.020$, range 0.271), where the matched constant-overlap run arced to 0.844 — and the energy-peak positions scatter across the whole window (frames [38, 74, 42, 10, 107, 128, 99, 69, 64, 28]; mode covers 1/10 within ± 1). The audio-SLB restoration shows directly in the boundary column: boundary similarity rises from 0.619 at the first hand-off to a 0.87–0.96 band from chunk 3 onward (first-to-last $0.619 \rightarrow 0.918$), against the 0.07–0.53 of the SLB-off configuration, where every chunk re-invented its audio. The SLB freeze is verified at every chunk (`SLB honored`= 1.0000).

6.3.4 Stage 2 Results

Stage 2 refines the 2×-upsampled latent in four overlapping windows (fpc = 40, overlap = 13, auto-selected from the 42k-token VRAM budget) against frozen Stage-1 audio (Section 5.2).

Table 9: I2V Stage 2 per-window metrics (4 windows). Fid is video fidelity to the Stage-1 latent (first / last frame of the window); Aud_bnd is the frozen Stage-1 audio's own continuity at the window seam; $g_{\mathrm{lo}}/g_{\mathrm{hi}}$ are the per-window detail-anchor gains.

Window	Bnd sim↑	Intra sim	Fid first	Fid last	Aud_bnd	$g_{\mathrm{lo}}/g_{\mathrm{hi}}$
1/4	N/A	0.7276	0.9851	0.9589	N/A	1.039/0.900
2/4	0.8285	0.7634	0.9546	0.9201	0.8588	1.044/0.900
3/4	0.7604	0.6768	0.9060	0.9699	0.7521	1.058/0.900
4/4	0.8961	0.6833	0.9425	0.9105	0.0513	1.049/0.900

End-of-window fidelity to the Stage-1 latent holds 0.91–0.97 across all four windows (fid_first 0.985 $\rightarrow$ 0.942, fid_last 0.959 $\rightarrow$ 0.910 first-to-last) — no end-of-run erosion. As in T2V, the distilled refinement adds spatial high-frequency energy at every window (pre-anchor hf share 0.45–0.48 against Stage-1 references of 0.34–0.36) and the per-window detail anchor pulls it back with g_{hi} at the 0.90 floor cap throughout; because each window is referenced to its own Stage-1 slice, the trim cannot accumulate into progressive softening. Video seam continuity ranges 0.76–0.90, weakest at the window-2→3 seam ($t \approx 26.7$ s). The audio timeline is bit-identical to Stage 1 (`frozen_verify_sim`= 1.0 per window); the Aud_bnd column re-measures the Stage-1 audio's own continuity where the video windows happen to cut. Two of the three readings are strong (0.859/0.752); the last window's 0.051 is the Stage-1 audio's own weak transition at $t = 40.0$ s surfacing verbatim in the frozen passthrough — Stage 2 never re-rolls audio, so it can neither cause nor repair it (the Stage-1 trend summary flags the corresponding span, Table 8). The assembled refined latents are $[1, 128, 160, 22, 40]$ (video, std 0.916) and $[1, 8, 1333, 16]$ (audio, frozen), with no NaN/Inf.

6.4 Computational Overhead

Table 10: Wall-clock timing on a 16 GB GPU (RTX-class, CPU block-streaming); s/it ranges span the matched T2V and I2V runs, which agree to within a few percent at every position. Stage 1 runs three transformer passes per step (split CFG, STG); Stage 2 runs a single video refinement pass per window (audio frozen). The Stage-1 window length varies per chunk because of the overlap jitter (Section 5.4), and the per-chunk cost tracks it.

Stage / pass	Chunks	Window (lf)	s/it	Time	Tokens
S1 (20 steps)	1	16	19.0	$\approx$ 6.3 min	—
S1 (20 steps)	2, 5, 8	20	21.5–23.1	$\approx$ 7.1–7.7 min each	—
S1 (20 steps)	3, 6, 9	22	23.6–23.9	$\approx$ 7.9 min each	—
S1 (20 steps)	4, 7, 10	24	25.0–26.1	$\approx$ 8.3–8.7 min each	—
S2 video (3 steps)	1/4	40	69–70	$\approx$ 3.5 min	35,200
S2 video (3 steps)	2–4/4	53	104–106	$\approx$ 5.2 min each	46,640

Total generation time: Stage 1 $\approx$ 77–78 min + Stage 2 $\approx$ 19 min, i.e. $\approx$ **96 min** (T2V) to $\approx$ **97 min** (I2V) for a $\approx$ 53-second clip. The dominant cost driver is the three-pass guidance in Stage 1 — each denoising step runs three transformer passes (conditional, unconditional, STG-perturbed) rather than one. The per-chunk cost now varies with the jittered window size (Section 5.4): chunk 1 is cheapest (19.0 s/it, 16 latent frames, no overlap context), and the continuation chunks step up cleanly with window length — 21.5–23.1 s/it at 20 lf, 23.6–23.9 at 22 lf, 25.0–26.1 at 24 lf — a per-chunk timing signature of the jitter itself. In Stage 2 the per-window cost rises $\approx$ 50% after window 1 (69–70 $\rightarrow$ 104–106 s/it) against a token-count rise of $\approx$ 32%: as the run proceeds an increasing share of the model is pushed to CPU (offloaded weights grow from $\approx$ 5.7 GB to $\approx$ 7.8 GB across the pass), so the slowdown is disproportionate to the token increase and is an offload-pressure effect rather than a compute-scaling one.

The T2V run costs the same as I2V to within rounding: per-chunk and per-window s/it match to a few percent at every position (e.g. Stage-1 chunk 1: 18.97 s/it in both runs; Stage-2 window 2: 103.8 vs. 106.0 s/it), confirming that the guide image changes neither the token count nor the compute cost — only the conditioning. The Stage-2 offload-pressure slowdown reproduces at the same windows in both runs, supporting that it is a property of the refinement pipeline rather than of run content.

7 Discussion

7.1 Effect of Chunk Length and Content Regime

Both runs used new_lf = 16 ($\approx$ 5.3 s per chunk), within the 21-frame safe limit established in earlier open-loop testing, streamed as ten chunks under the active guidance stack, with the overlap jitter (Section 5.4) varying the context length per chunk (4–8 lf). In neither run does boundary similarity degrade with chunk index. I2V *improves* monotonically in the large: its floor is the first SLB-conditioned hand-off (0.7900, chunk 2) and its strongest seam is the last (0.9814). T2V is flat first-to-last (0.796 $\rightarrow$ 0.790) with the weakest seams mid-run (0.469 at chunk 5, 0.541 at chunk 6) rather than late: where continuity is weak, it is weak as isolated seam variance — chunk 5 is the first half-overlap (4 lf) window following a full-overlap chunk — not as compounding drift. The two runs exercise opposite content regimes — I2V a stable anchored scene (intra-chunk similarity plateauing at 0.82–0.91), T2V sustained motion (intra 0.02–0.71 throughout) — and the correction stack scales with the measured excess: near-idle on I2V (all band gains $\geq$ 0.983, detail-anchor gains within +4%/−3% of unity) versus working measurably harder on T2V (band-2 gains to 0.9636, g_{hi} down to 0.918 against the raw high-band overshoot).

7.2 Identity Tracking and Boundary Self-Consistency

Chunk 2's identity_sim (0.7900) equals its boundary_sim exactly. This is not a coincidence but a *self-consistency check*: the anchor was seeded from chunk 1's last frame, and the SLB passes that same frame as the first latent frame of chunk 2's conditioning, so the anchor query and the boundary query compare the same source frame (chunk 1 frame 15) against the same target frame (chunk 2 frame 0). Both metrics agreeing validates that the anchor bank and the SLB operate on the same reference, and that the τ_c re-noising at the boundary introduces only minimal distributional shift. For later chunks the bank holds multiple entries and the identity query selects a different anchor than the boundary, so the two metrics diverge, as expected (e.g. I2V chunk 6: identity 0.878 vs boundary 0.940). The same self-consistency check holds in the unrelated T2V run at the same chunk (identity 0.7957 = boundary 0.7957, Table 2), confirming the mechanism is a property of the anchor/SLB bookkeeping rather than of any particular content.

7.3 Audio Coherence and the Role of Visual Context

A natural hypothesis is that a guide image shapes streaming audio behaviour: in LTX-2.3's joint audio-video transformer, video and audio tokens share self-attention layers, so a strong single-frame visual prior could constrain audio generation implicitly through cross-modal attention — for better (a coherence anchor) or worse (a repetition template). The matched pair does not support the hypothesis in either direction. Chunk-1 audio coherence differs without any guide-image effect to attribute it to (T2V $0.745 \rightarrow 0.622$ vs. I2V $0.347 \rightarrow 0.523$), and with the envelope repetition now eliminated in *both* modes by the jitter (env_corr ≤ -0.003 throughout in T2V, ≤ 0.190 in I2V; Section 5.4), there is no repetition signature left for the guide image to modulate either. The remaining audio differences track the prompted *sound content* — T2V's loud continuous ambience (RMS reference 0.8404) naturally self-correlates within and across chunks, while I2V's quieter, more varied scene audio (RMS reference 0.5589) does not — rather than the presence or absence of a visual prior. The practical conclusion is negative but useful: visual conditioning is not a lever on streaming audio behaviour at this scale, and the audio failure modes of Section 7.6 had to be — and were — addressed in the audio path itself.

7.4 Stability Margin

With $\rho_{\text{loop}} = 0.570$, the CLSS feedback loop provides a $\approx 43\%$ gain reduction per chunk before accounting for the model's open-loop gain g. In the I2V run the spectral-shrinkage correction was light throughout — band-2/3 gains of 0.983–0.998, a ≈ 1–2% high-frequency energy excess over the EMA reference. Taking the typical ≈ 1–2% per-chunk growth as a rough estimate of the open-loop gain in the spectral channel gives $g \approx 1.02$ and $\rho_{\text{closed}} \approx 0.57 \times 1.02 \approx 0.58$, comfortably below the stability boundary of 1. The T2V run runs hotter on this channel: band-2 gains fall to 0.964–0.985 for most chunks, a ≈ 2–4% typical excess, giving $g \approx 1.04$ and $\rho_{\text{closed}} \approx 0.57 \times 1.04 \approx 0.59$ — still comfortably stable, and close enough to the I2V estimate that the margin appears to be a property of the correction stack rather than of the conditioning mode. The raw audio spectral channel, not covered by the video-side machinery of Sections 3.3 and 3.5, is much improved over the pre-jitter runs but remains the least-corrected one: the Stage-1 trend summary's `aud_hf` statistic (high-band audio energy relative to the chunk-1 reference) now *declines* in T2V ($0.885 \to 0.772$; the residual energy excess concentrates in the lowest frequency bin instead, up to $1.62\times$ reference) and grows only mildly in I2V ($0.881 \to 1.047$, range 0.360, with top bins individually reaching $\approx 1.9\times$ reference at chunk 5) — against the unbounded $0.42 \to 1.33$ (T2V) and $0.87 \to 1.28$ (I2V) growth of the constant-overlap, SLB-off runs. The RMS anchor constrains only global loudness; no current correction targets the audio frequency distribution (Section 7.6).

7.5 Stage-2 Identity Metric Ambiguity

Stage 2 lacks an anchor bank; its identity_sim reduces to comparing each window against the fixed first-window reference. On a single moving scene this conflates ordinary content motion with drift: in I2V, where the reference is the guide-image region, windows 2–4 sit at a 0.51–0.53 plateau, reflecting that the video has moved away from its opening frames, not instability. In T2V, where window 1 has no anchor to begin with, the same comparison instead oscillates around zero (-0.36 to $+0.06$ across windows 2–4) — consistent with window 1 itself being internally uncorrelated (Table 5, intra -0.322), so there is no stable reference for later windows to be "close to" or "far from" in the first place. Adding a scene-aware anchor bank to Stage 2 would make this metric interpretable in either conditioning mode; as reported it should be read only as "distance from the opening," not as a drift signal.

7.6 Measured Failure Modes at Longer Horizons

Per-frame telemetry across extended runs — including the ten-chunk, ≈ 53-second T2V and I2V runs reported above and earlier 13-lf and 31-lf runs — exposed two repetition mechanisms with distinct signatures. One is now diagnosed and solved; the other remains open.

Loudness-envelope repetition: diagnosed and solved. Earlier Stage-1 configurations repeated a per-chunk loudness gesture: env_corr crossed the 0.7 repetition flag in both conditioning modes (sustained 0.76–0.96 in T2V; an arc to 0.844 in I2V), it was the flagged drift of both trend summaries, and the repetition was audible in listening across every configuration tested. Every intervention available at the conditioning and guidance level was tested live against it, and *all failed*: enabling the audio SLB under a constant overlap (repetition unchanged, plus high-frequency runaway $1.02 \to 1.51$ as the buffer froze each chunk's loud tail forward); raising modality guidance to the reference value (repetition unchanged, plus audible hiss/drone as guidance energy rose); and perturbing the reference-audio window itself — blending in noise, decaying its length over chunks, flattening its loudness envelope (repetition unchanged in each case, with the energy-injecting variants adding hiss). Those failures are retained here as the diagnostic they turned out to be: if no change to conditioning *values* or guidance *weights* touches the artefact, the artefact must live in the one thing none of the interventions changed — the *shape* of the chunk window.

With a constant overlap, every continuation chunk denoises an identical window and keeps the same phase of the model's window arc, so the arc is a fixed point of the chunk map (Section 5.4); the lock engages when the reference-audio window reaches full length at chunk 2–3, matching the user-audible "audio repeats after chunk 1". The per-chunk overlap phase-jitter breaks the fixed point by construction, and with the lock gone the audio SLB is safe to restore as the content-continuity carrier. In the reported runs the signature is absent: env_corr never exceeds -0.003 (T2V) or 0.190 (I2V) at any hand-off, the peak-phase detector scatters (1/10 mode coverage in both runs), boundary similarity holds a 0.85–0.92 typical band with the SLB honored at 1.0000 throughout, and the restored SLB shows no runaway (T2V `aud_hf` 0.885 $\rightarrow$ 0.772). A full-video single-pass audio re-render was also implemented and worked for short clips, but was rejected by design — CLSS exists for arbitrary-length streaming, and a whole-video window defeats it (Section 5.4).

Chunk-relative phase lock, broken by construction. An earlier symptom of the same mechanism was a hard *phase* lock: the audio energy peak at 94% of the window for eight consecutive 13-lf chunks, and looser but persistent clustering under constant-overlap 16-lf runs. With the jitter, no phase can recur: the detector reads fully scattered in both reported runs (audio peak mode covering 1/10 chunks within ± 1; T2V peaks [38, 118, 127, 43, 101, 64, 26, 86, 49, 12], I2V peaks [38, 74, 42, 10, 107, 128, 99, 69, 64, 28]), and env_corr confirms the gesture itself is gone, not merely re-phased.

A $\sim$4-second video layout oscillation. Coarse scene layout departs from the scene reference and partially returns with a $\approx 4\,$s wall-clock period that is *invariant to chunk length*: at 13-lf (4.3 s) chunks the dip lands at new-frame 5–7 in every chunk (appearing seam-locked), while at 31-lf (9.9 s) chunks it recurs within the chunk at the same seconds-period. Context strength and context length (overlap 3/5/8) were tested and neither removes it; the oscillation appears intrinsic to the model's trained temporal horizon. In the reported 16-lf jittered runs the layout-argmin detector reads unlocked in both modes (T2V mode 4 covering 5/10 chunks within ± 1, I2V mode 12 covering 4/10) — the strict per-chunk swing-and-return does not phase-lock to the seam grid, though global layout still wanders (largest per-frame layout drops of -0.67 in both runs, spread across the run rather than concentrated at one seam). This failure mode remains open.

Stage-2 audio is frozen by design. The freeze (Section 5.2) is the outcome of a measured escalation. With per-window audio re-rendering at high denoise (σ-scale 1.0), $\approx 45\%$ of the audio is re-rolled per window and the re-rolls disagree at every window boundary (boundary cosine 0.32–0.42) *regardless* of the audio SLB length — the SLB pins the seam frame, not the style of the re-roll. At denoise 0.4 the pass polishes instead (seed similarity 0.79–0.81) yet seams still disagreed at 0.59–0.72: polishing narrows the disagreement but cannot close it while windows are denoised independently. Freezing (mask 0) closes it exactly — Stage 2 introduces no audio seams at all, and the only audio discontinuities in the final output are Stage 1's own chunk boundaries. Rule: audio is either generated in one window or not re-rolled at all; any multi-window audio denoise re-seams the timeline it was meant to improve.

Audio high-frequency growth, mostly closed. The trend-summary's `aud_hf` statistic (high-band audio energy relative to the chunk-1 reference) grew unboundedly in the pre-jitter runs ($0.42 \to 1.33$ T2V, $0.87 \to 1.28$ I2V), and under a constant overlap the audio SLB made it worse ($1.02 \to 1.51$). In the reported jittered, SLB-on runs the channel is closed in T2V ($0.885 \to 0.772$) and reduced to a mild residual in I2V ($0.881 \to 1.047$, range 0.360; top bins to $\approx 1.9\times$ reference at chunk 5). No current correction targets this channel directly: the video-side spectral shrinkage (Section 3.3) and detail anchor (Section 3.5) operate on the video latent only, and the audio-side RMS anchor (Section 5.3) constrains global loudness but not the frequency distribution. An audio-side two-band anchor, mirroring Section 3.5 but applied to the audio latent's frequency axis, is the natural next mechanism to try; an earlier attempt at audio high-band shrinkage was reverted for audible artefacts, so any such correction must be validated by ear, not by the statistic it improves.

8 Limitations

Single hyperparameter configuration, single seed. The reported T2V and I2V results share one set of hyperparameters (new_lf = 16, overlap = 8, 10 chunks, 1 scene), one seed, and one hardware configuration (16 GB VRAM, RTX-class); they differ only in the first-frame conditioning, by design, to isolate that variable. No baseline comparison against open-loop streaming is reported, and no ablation of the individual CLSS components (re-noising alone, AdaIN alone, shrinkage alone, anchor bank alone) or of the individual guidance terms (split CFG, modality, STG) has been performed. The paper therefore cannot quantify the marginal contribution of each correction or guidance term, or demonstrate improvement over the open-loop baseline from the same metrics, and every single-run comparison in Section 7 (e.g. the T2V/I2V content-regime contrast) is a $n = 1$-per-arm observation, not a statistically powered result. These comparisons are left to future work.

Stability metrics vs. perceptual quality. The reported metrics—boundary cosine similarity, intra-chunk cosine similarity, audio within-chunk segment similarity—measure the *stability* of the latent representation across chunks. They do not assess the perceptual quality of the generated content, prompt adherence, or realism. Standard quality metrics (FID, IS, CLIP similarity, human evaluation) would be required to characterise the generated video as a content generation system. High boundary similarity is a necessary but not sufficient condition for high video quality.

Object-level identity is unobserved. Every reported metric is computed from mean-pooled channel features or low-band spatial maps and is therefore *object-blind*: a vehicle or person can be replaced by a different instance across a chunk boundary while boundary and identity cosines remain above 0.95. This failure class was observed perceptually and is invisible to the telemetry. High-similarity readings must not be interpreted as object persistence. The available mitigation is textual: one prompt per chunk via `CLSSScenePrompts`, re-describing the persistent objects explicitly. Object-level tracking metrics (detection/re-identification on decoded frames) are future work.

Stability model assumptions; g_{SLB} not yet measured. The linear gain model in Section 3.6 assumes that g is approximately constant and that the AdaIN and re-noising corrections act as independent multiplicative attenuators. Both assumptions simplify the analysis: the actual closed-loop dynamics involve higher-order interactions between corrections, guidance, and scene content. A more rigorous treatment (e.g., Lyapunov stability for the latent iteration, or a Monte Carlo sweep over perturbation seeds and chunk positions) would require characterising the denoiser's Jacobian across a distribution of latent inputs, and is left to future work. Concretely: the g_{SLB} perturbation protocol (Eq. 10) is implemented and verified at the tensor level — the perturbation is confined exactly to the overlap region, and the norm-ratio formula recovers a known injected gain in a synthetic test — but it doubles per-chunk compute, so it is compiled off in the node (no exposed toggle) and has not been exercised against a live transformer. Neither reported experiment used it: the $g \approx 1.02$–1.05 figures in Section 7 remain an informal band-energy proxy, not this measurement. Re-enabling it in code for a dedicated measurement run is the direct next step toward an empirically grounded, rather than assumed, value of g.

No working anchor-bank conditioning. The anchor bank tracks long-range identity but does not condition generation: both wirings attempted to date have failed. The reference library's append-style `VideoConditionByKeyframeIndex` reproduces the AV audio corruption already diagnosed for the i2v guide path (Section 5.6). A low-strength in-place replacement at the first new frame — verified in isolation to place the anchor correctly and leave the SLB region untouched — was exercised in a full run and produced visible content *morphing*: it injected a past keyframe into every chunk, and because retrieval returns the least-similar non-redundant anchor, it dragged each chunk back toward content the scene had already moved past (one chunk was pulled to an earlier anchor than a previous chunk had used). It was reverted. Conditioning the joint audio-video model on a temporally-displaced keyframe without either corrupting audio (append) or morphing video (in-place replace) is an unsolved design problem; the $\sim$4-second video layout oscillation (Section 7.6) that motivated the attempt remains unaddressed.

Phase-jitter caveats. The overlap jitter (Section 5.4) is validated at the ten-chunk, $\approx$ 53-second horizon. (a) Half-overlap chunks carry only 4 lf of context: no video-continuity loss attributable to the shortened context is measured at this horizon (the T2V mid-run seam floor of 0.469 recovers on the following chunks and the trend is flat). Longer horizons are simply not tested yet — the limiting factor is the hardware budget available for this work (each $\approx$ 53-second clip costs $\approx$96–97 min on a single

16 GB GPU), not any known property of the method; longer runs are planned once better hardware is available. (b) The jitter cycle is fixed ([full, half, $\frac{3}{4}$]), not content-adaptive; a schedule that reacts to measured envelope correlation is unexplored. (c) I2V audio high-frequency energy still grows mildly over the run ($0.881 \rightarrow 1.047$; Section 7.6). (d) The I2V Stage-2 last-window `aud_bnd` of 0.051 is the Stage-1 audio's own weak transition at $t = 40.0$ s surfacing verbatim in the frozen passthrough — a limitation of the Stage-1 audio at that span, not of the refinement pass, which never re-rolls audio.

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes (e.g., Q8 or FP8).

9 Related Work

Long video generation. StreamingT2V [3] and FIFO-Diffusion [4] generate long videos by sliding-window denoising. Gen-L-Video [5] uses temporal attention patching. FreeNoise [1] reschedules the initial noise (window-based fusion of a shuffled, reused noise sequence) to extend a pretrained model's temporal attention window without retraining or sliding-window denoising at all — a tuning-free approach to the same underlying problem (fixed training-time context length) that CLSS addresses by chunking with closed-loop correction instead. VidToMe [2] merges self-attention tokens across frames to improve temporal consistency in zero-shot video editing, operating within a single generation pass rather than across autoregressively-generated chunks. CLSS differs from all of these by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models, where each chunk is a genuinely separate denoising process conditioned on the previous chunk's output — the exposure-bias setting FreeNoise and VidToMe do not target.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [6]. For diffusion models, Ning et al. [7] and Salimans & Ho [8] address related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [9] was introduced for style transfer, where the goal is to match the mean and standard deviation of a content feature map to those of a style reference. EMA-based statistical normalisation across video frames has also appeared in temporal consistency methods. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [10] and Ceylan et al. [11] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

10 Conclusion

We presented CLSS, a five-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The corrections — calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor — achieve $\rho_{\text{loop}} = 0.570$; driven by a reference-parity multi-modal guider (split per-modality CFG and STG) that correctly handles ComfyUI's packed audio-video latent representation, the pipeline held Stage 1 video boundary similarity on a rising trajectory ($0.79 \rightarrow 0.98$ I2V) across ten chunks of matched ≈ 53-second single-scene clips, held the high-frequency energy share flat end-to-end in both conditioning modes, and held audio loudness to its chunk-1 reference within $\pm 5\%$ throughout. A two-stage pipeline (streaming Stage 1 + windowed distilled Stage 2 with per-window detail anchoring and frozen audio) delivers a fully refined $2\times$-resolution audio-video output in $\approx$ 96–97 minutes per clip on a single 16 GB GPU, essentially independent of conditioning mode.

The headline audio result is that the per-chunk loudness-envelope repetition — the metronome that earlier versions of this paper characterised as structural and immune to intervention — is diagnosed and solved. The diagnosis: the repeated gesture is a *fixed point of the constant-shape chunk grid*, not a property of any conditioning value, which is why every conditioning- and guidance-level intervention failed against it. The fix is chunk-native: a per-chunk overlap phase-jitter (cycling [full, half, $\frac{3}{4}$] of the configured overlap, so no two consecutive chunks share a window shape) breaks the fixed point by construction, and with the phase lock gone the audio SLB is restored as the content-continuity carrier and the RMS anchor keeps energy stability. Against the matched pre-jitter runs, envelope correlation drops from a sustained 0.76–0.96 to ≤ 0.19 at every hand-off (-0.51–0.19 across both runs), audio boundary similarity rises from 0.07–0.53 to a 0.85–0.92 typical band with the SLB verified honored at every chunk, phase-lock detectors read fully scattered (1/10 mode coverage in both runs), and audio high-frequency growth is closed in T2V ($0.885 \rightarrow 0.772$) with a mild residual in I2V ($0.881 \rightarrow 1.047$) — all while preserving arbitrary-length streaming, an unchanged output timeline, and exact audio/video proportionality.

Running matched T2V and I2V generations under an identical seed and guidance stack, differing only in the presence of a guide image, shows that visual conditioning is not a lever on streaming audio behaviour: chunk-1 audio coherence, envelope statistics, and loudness stability all track the prompted sound content rather than the guide image, while the guide image's clear effects are on the video side (a stable anchored scene with a rising $0.79 \rightarrow 0.98$ boundary trajectory versus T2V's variable 0.47–0.92 seams under sustained motion). One repetition mechanism remains open at this horizon: a chunk-length-invariant $\approx$ 4-second video layout oscillation that appears intrinsic to the model's trained temporal window (its detector reads unlocked in both reported runs, but global layout still wanders). All of this is visible in the pipeline's per-chunk telemetry — unconditional settings dumps, drift-event tables, and a phase-lock detector — which is designed to localise such effects in wall-clock time and latent-frame position, and to make a silently inactive guidance path impossible to miss (a guidance term that does not change per-step wall-clock time is not running).

Object-level identity persistence across chunk boundaries, invisible to the current structural metrics in both conditioning modes, remains the principal open problem for content-level quality.

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